

RDROP / 2-LAYER LOW-VOLTAGE CONTROLLER
External HV regulators / 7,4 V or 5 V servo / ESP32 / AS5600 / dual optical gates
Regulated inputs only. Assembly A: F101 fitted, F102 DNP. Review report before fabrication.

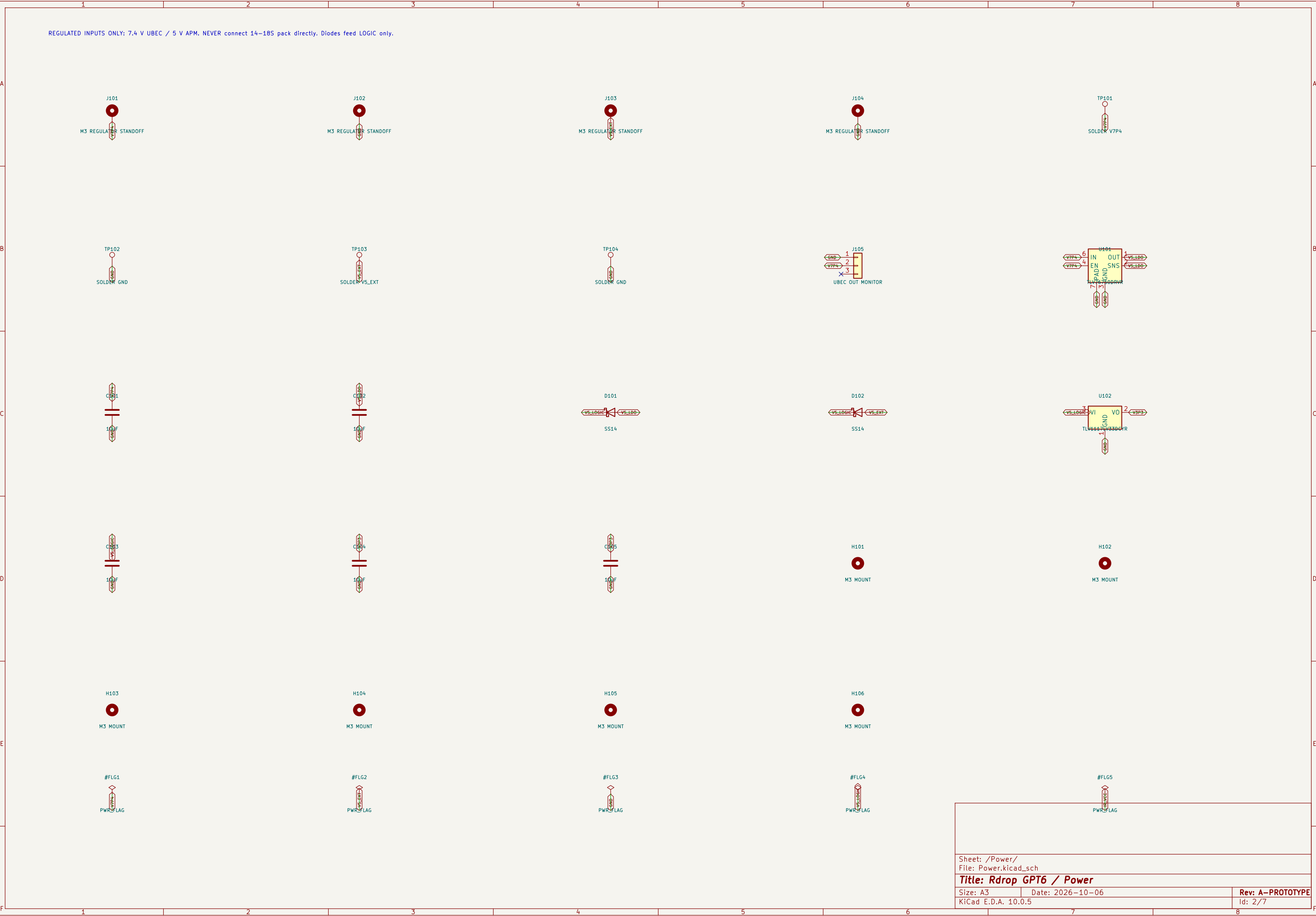
Block	File
Power	File: Power.kicad_sch
Servo	File: Servo.kicad_sch
MCU	File: MCU.kicad_sch
Interface	File: Interface.kicad_sch
Magnetic	File: Magnetic.kicad_sch
Optical	File: Optical.kicad_sch

Sheet: / File: Rdrop-GPT6.kicad_sch		
Title: Rdrop GPT6 / System		
Size: A4	Date: 2026-10-06	Rev: A-PROTOTYPE
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3.3 V logic only. Hold BOOT, tap RESET for UART download. GPIO34/35 are input-only. No servo current through logic rails.

The schematic shows an ESP32-WROOM-32E-N4 module (U201) with the following components and connections:

- Capacitors:**
 - CAP1, CAP2, CAP3, CAP4: 10µF electrolytic capacitors connected to VDD and GND.
 - CAP5: 100k resistor connected to VDD and GND.
- Resistors:**
 - R01, R02, R03, R04: 10k resistors connected to various pins and GND.
- Switches:**
 - SW201: RESET switch connected to the RESET pin.
 - SW202: BOOT switch connected to the BOOT pin.
- Other Components:**
 - U201: ESP32-WROOM-32E-N4 module.
 - U202: 100k resistor connected to VDD and GND.

The module pins are labeled as follows:

- EN
- SENSOR_VP
- SENSOR_VN
- TXD0/I01
- RXD0/I03
- I04
- I05
- I012
- I013
- I014
- I015
- I016
- I017
- I018
- I019
- I020
- I021
- I022
- I023
- I025
- I026
- I027
- I032
- I033
- I034
- I035

Title: Rdrdp GPT6 / MCU		
Size: A3	Date: 2026-10-06	Rev: A-PROTOTYPE
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